

Credit Card Users Churn Prediction

Project 3 and Advanced Machine Learning

Date 24.10.2024

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- Model performance summary for hyperparameter tuning.
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Executive Summary

- Actionable insights & recommendations

Based on the analysis of customer data, here are some key insights and recommendations to improve customer retention and mitigate churn:

1. Focus on Customer Satisfaction:

- **Address pain points:** Identify and address specific pain points that may be driving customers to leave, such as high fees, poor customer service, or limited card benefits.
- **Improve customer service:** Enhance customer support channels and ensure prompt and efficient responses to inquiries and complaints.
- **Personalize offerings:** Tailor products and services to individual customer needs and preferences.

2. Review Fee Structure:

- **Evaluate fee fairness:** Assess the fairness and competitiveness of fees charged to customers.
- **Consider alternative fee structures:** Explore options to reduce or eliminate certain fees, or offer value-added benefits to offset them.
- **Provide transparency:** Clearly communicate fee details to customers, ensuring they understand the charges and how they are calculated.

3. Enhance Product Offerings:

- **Expand product range:** Offer a wider variety of credit cards to cater to different customer segments and needs.
- **Introduce innovative features:** Provide unique and attractive features that differentiate Thera Bank's cards from competitors.
- **Reward loyalty:** Implement loyalty programs or rewards systems to incentivize customer retention.

4. Improve Customer Engagement:

- **Personalized communication:** Send targeted messages and offers based on customer preferences and behavior.
- **Leverage digital channels:** Utilize digital channels like email, SMS, and social media to engage with customers effectively.
- **Encourage feedback:** Seek customer feedback through surveys and other channels to understand their needs and concerns.

5. Monitor and Analyze Customer Behavior:

- **Track key metrics:** Continuously monitor metrics such as customer lifetime value, transaction frequency, and spending patterns.
- **Identify at-risk customers:** Use predictive analytics to identify customers who are likely to churn and intervene proactively.
- **Refine strategies:** Adjust marketing and retention efforts based on ongoing analysis of customer behavior.

Business Problem Overview and Solution Approach

Problem statement

The Thera Bank is facing a decline in credit card usage. To identify customers at risk of churning, build a classification model using customer data. Analyze the model's output to understand the reasons for churn and develop strategies to improve customer retention and mitigate financial losses.

Solution Approach / methodology:

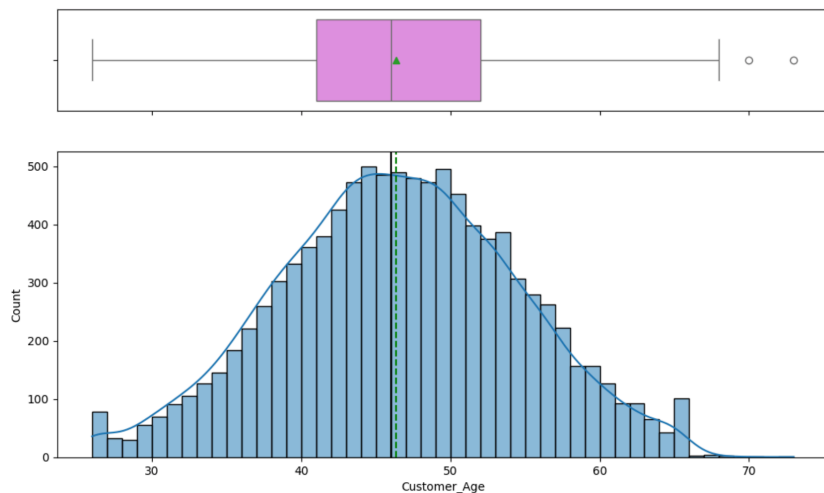
1. **Data Collection and Preparation:** Gather relevant customer data, including demographics, transaction history, credit card usage patterns, and customer interactions. Clean and preprocess the data to ensure accuracy and consistency.
2. **Feature Engineering:** Create new features or transform existing ones to capture meaningful patterns and relationships within the data. This might involve aggregating transaction data, calculating customer lifetime value, or normalizing numerical features.
3. **Model Selection and Training:** Choose a suitable classification algorithm (e.g., logistic regression, decision tree, random forest, or gradient boosting) and train it on the prepared dataset. Experiment with different hyperparameters to optimize model performance.

Model Evaluation: Assess the model's performance using appropriate metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. Consider cross-validation to evaluate the model's generalization ability.

Interpretation and Insights: Analyze the model's output to identify the most important features that contribute to churn. Use these insights to understand the underlying reasons for customer churn and develop targeted strategies to address them.

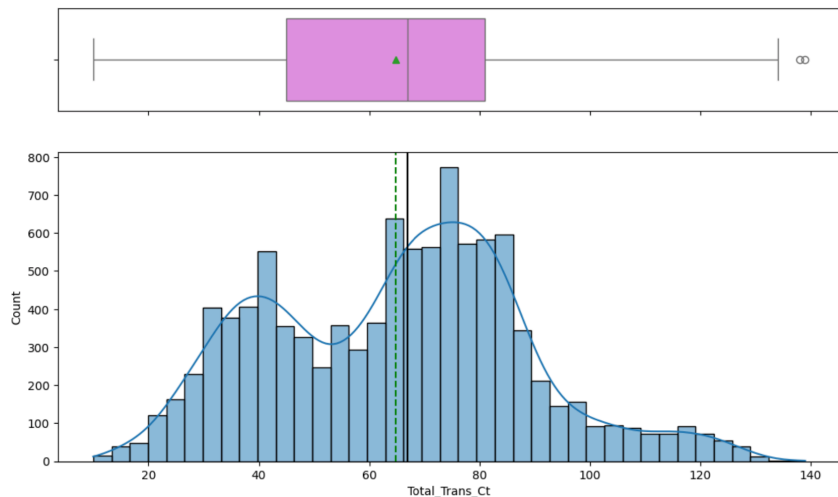
Deployment and Monitoring: Deploy the trained model into a production environment to predict churn for new customers. Continuously monitor the model's performance and retrain it as needed to adapt to changing conditions.

EDA Results



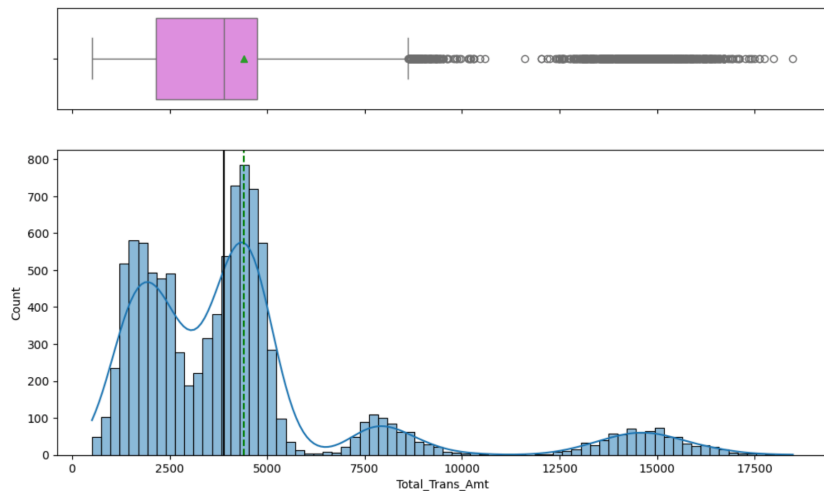
- Customer_Age
- The customer age distribution appears to be **approximately normally distributed**, with a peak around the mean age.
- There are some outliers, particularly on the higher end, indicating a few older customers.
- The median age is slightly lower than the mean age, suggesting a slight left skew in the distribution.
- This visualization provides a good overview of the age distribution in the dataset.

[Link to Appendix slide on data background check](#)



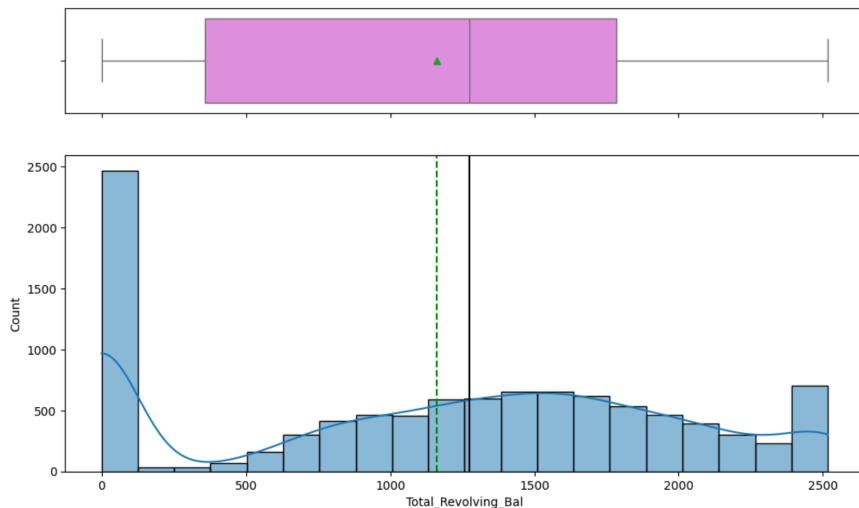
● Total_Trans_Ct

- The distribution of "Total_Trans_Ct" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with a significantly higher number of transactions compared to the majority.
- The median number of transactions is slightly lower than the mean, suggesting that the majority of customers have fewer transactions than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high transaction counts.
- The overall shape of the distribution suggests that most customers have a moderate number of transactions, with a smaller number of customers having either very few or very many transactions.



● Total_Trans_Amt

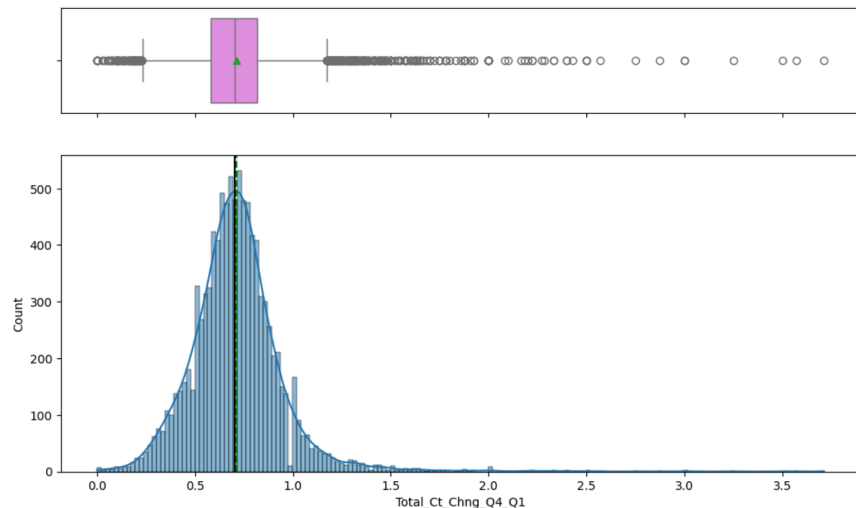
- The distribution of "Total_Trans_Amt" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher total transaction amounts compared to the majority.
- The median total transaction amount is slightly lower than the mean, suggesting that the majority of customers have lower total transaction amounts than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high total transaction amounts.
- The overall shape of the distribution suggests that most customers have a moderate total transaction amount, with a smaller number of customers having either very low or very high total transaction amounts.



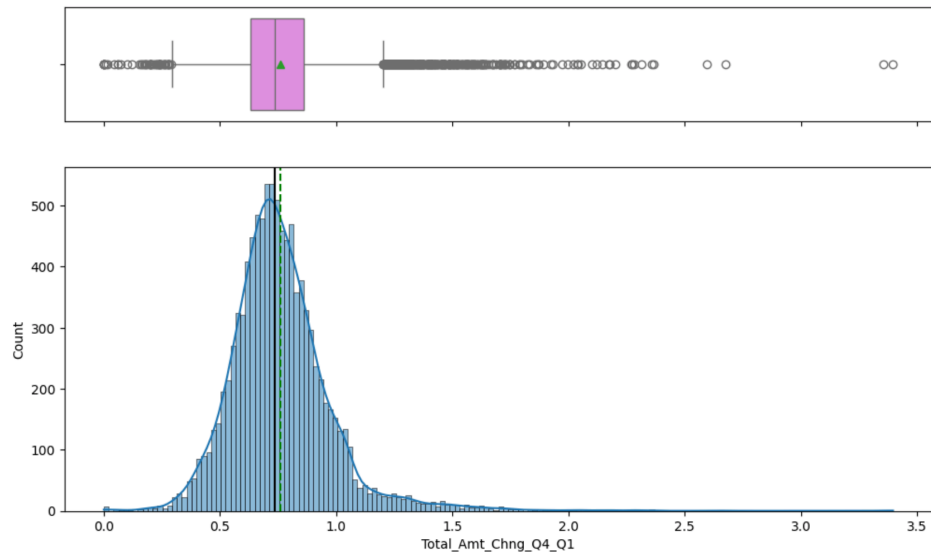
● Total_Revolving_Bal

- The distribution of "Total_Revolving_Bal" indicates that there are some customers with significantly higher total revolving balances compared to the majority.
- The median total revolving balance is slightly lower than the mean, suggesting that the majority of customers have lower total revolving balances than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high total revolving balances.
- The overall shape of the distribution suggests that most customers have a moderate total revolving balance, with a smaller number of customers having either very low or very high total revolving balances.

- Total_Ct_Chng_Q4_Q1

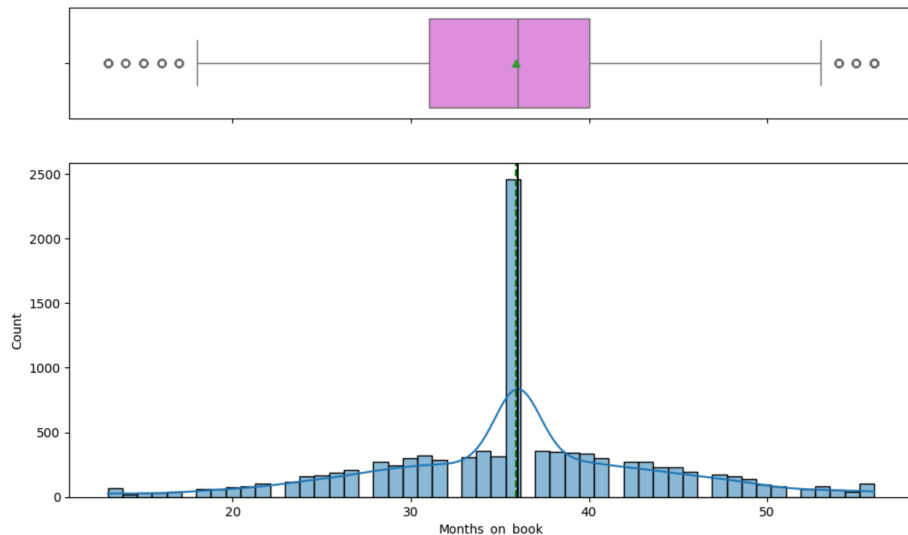


- The distribution of "Total_Ct_Chng_Q4_Q1" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher total count changes compared to the majority.
- The median total count change is slightly lower than the mean, suggesting that the majority of customers have lower total count changes than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high total count changes.
- The overall shape of the distribution suggests that most customers have a moderate total count change, with a smaller number of customers having either very low or very high total count changes.



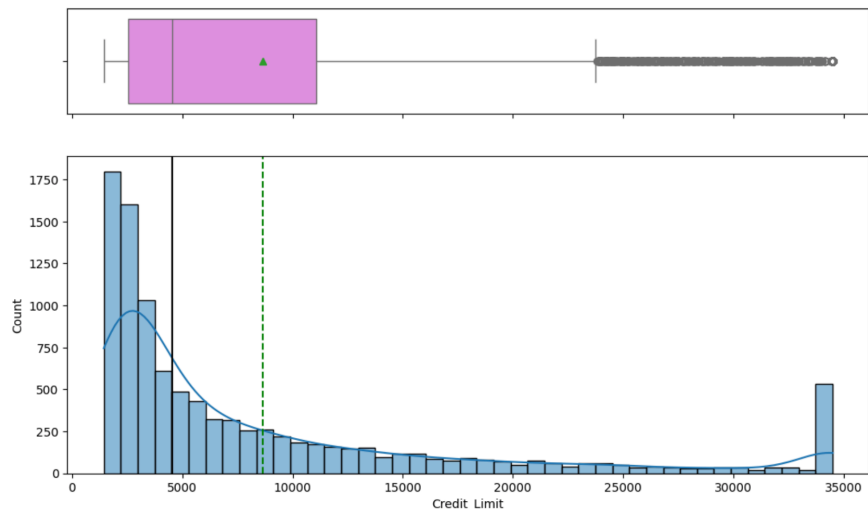
• Total_Amt_Chng_Q4_Q1

- The distribution of "Total_Amt_Chng_Q4_Q1" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher total amount changes compared to the majority.
- The median total amount change is slightly lower than the mean, suggesting that the majority of customers have lower total amount changes than the average.
- The overall shape of the distribution suggests that most customers have a moderate total amount change, with a smaller number of customers having either very low or very high total amount changes.



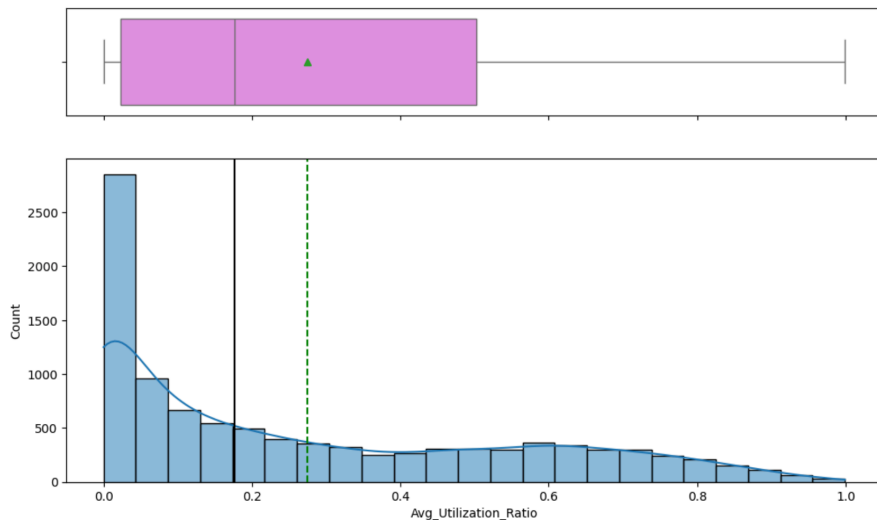
● Months_on_book

- The distribution of "Months_on_book" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher number of months on book compared to the majority.
- There are a few outliers on the higher end, indicating some customers who have been with the bank for an exceptionally long time.
- The overall shape of the distribution suggests that most customers have been with the bank for a moderate number of months, with a smaller number of customers having either very short or very long tenures.



● Credit_Limit

- The distribution of "Credit_Limit" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher credit limits compared to the majority.
- The median credit limit is slightly lower than the mean, suggesting that the majority of customers have lower credit limits than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high credit limits.
- The overall shape of the distribution suggests that most customers have a moderate credit limit, with a smaller number of customers having either very low or very high credit limits.



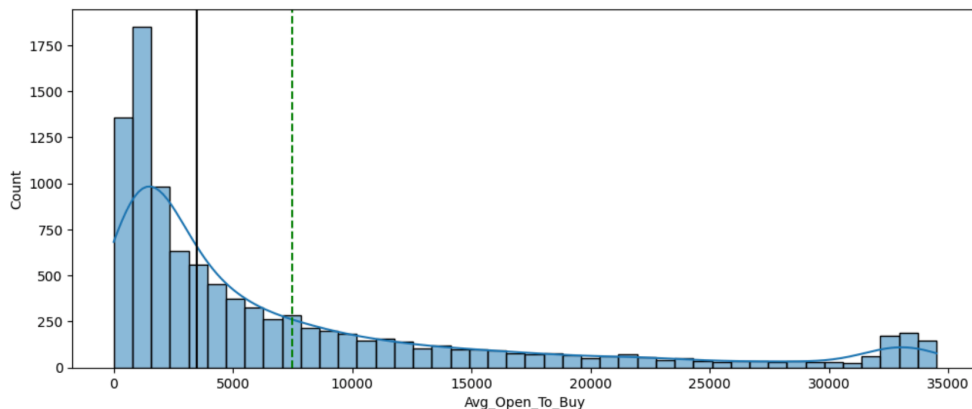
- Avg_Utilization_Ratio

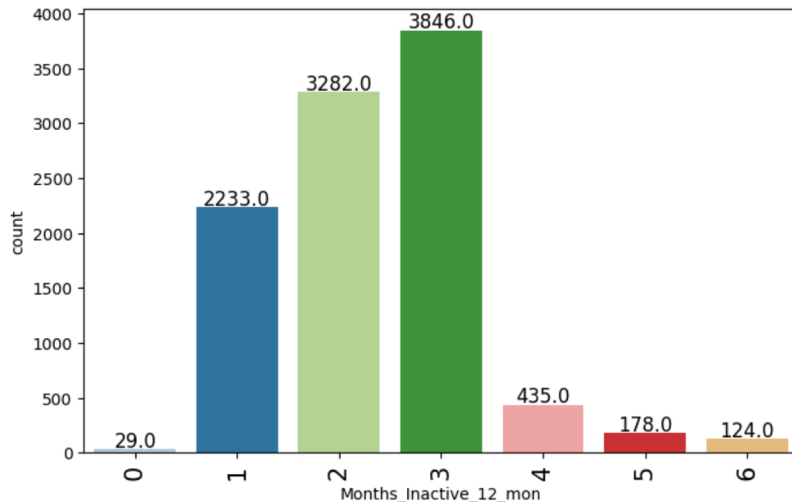
- The data for "Avg_Utilization_Ratio" is skewed to the right, with a longer tail on the right side. This indicates that there are some higher values that are more spread out compared to the lower values.
- The median (represented by the line inside the box) is closer to the lower end of the distribution, suggesting that the majority of values are on the lower side.
- The outlier points visible in the box plot indicate that there are some unusually high values of "Avg_Utilization_Ratio".



● Avg_Open_To_Buy

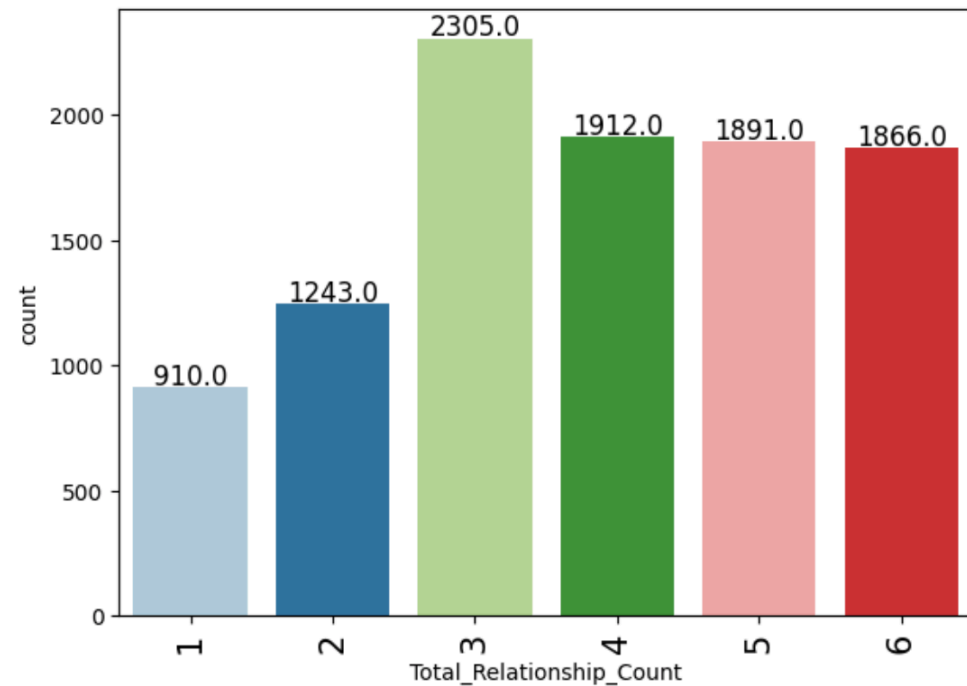
- The distribution of "Avg_Open_To_Buy" is **right-skewed**, with a longer tail on the right side. This indicates that there are some customers with significantly higher average open to buy compared to the majority.
- The median average open to buy is slightly lower than the mean, suggesting that the majority of customers have lower average open to buy than the average.
- There are a few outliers on the higher end, indicating some customers with exceptionally high average open to buy.
- The overall shape of the distribution suggests that most customers have a moderate average open to buy, with a smaller number of customers having either very low or very high average open to buy.





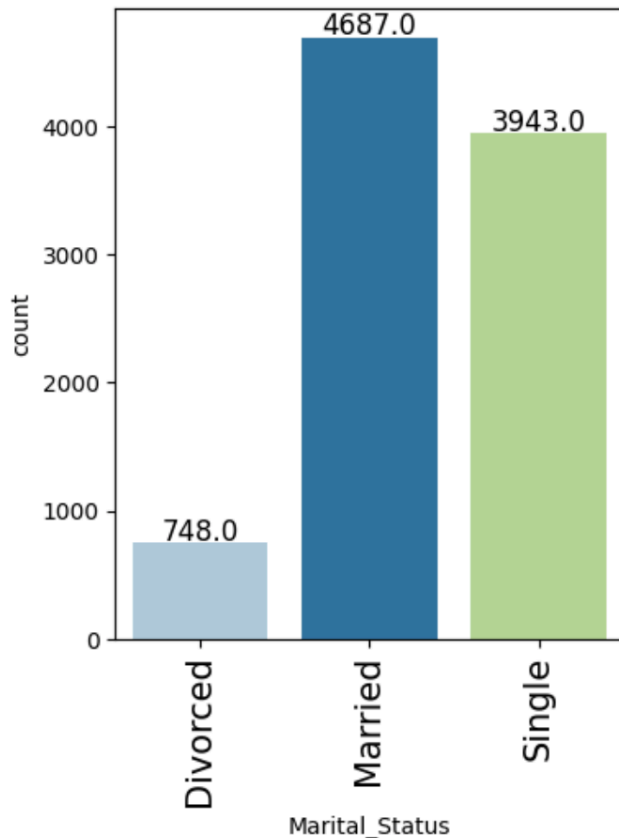
- Months_Inactive_12_mon

- The bar plot shows the distribution of the "Months_Inactive_12_mon" variable.
- The x-axis represents the number of months inactive, and the y-axis represents the count of customers.
- The chart indicates that most customers have been inactive for 0-3 months.
- There is a significant drop-off in the number of inactive customers after 3 months.
- The overall distribution suggests that most customers remain active for a short period before becoming inactive.



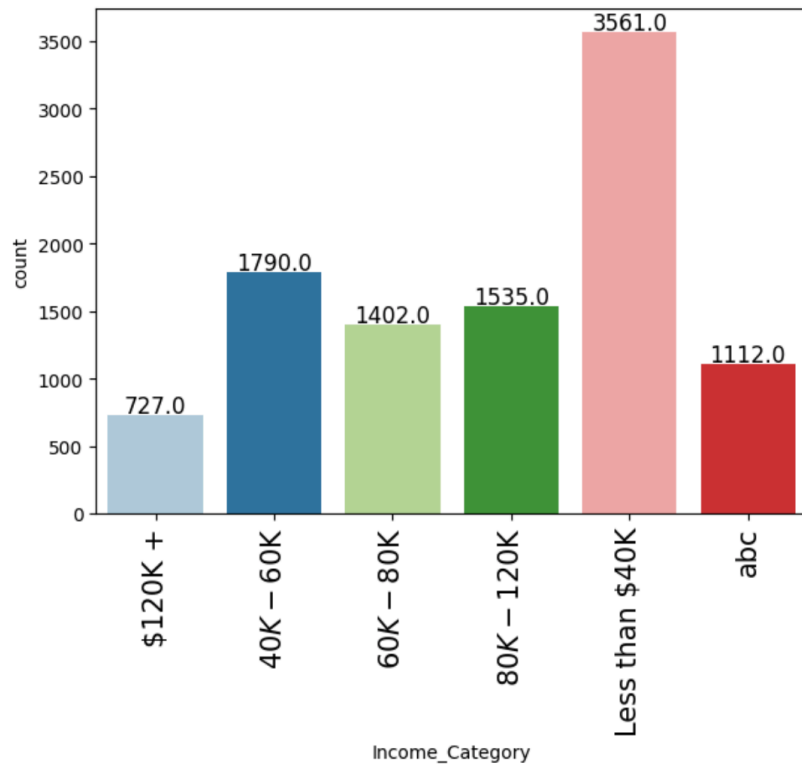
- Total_Relationship_Count

- The bar plot shows the distribution of the "Total_Relationship_Count" variable.
- The x-axis represents the total number of relationships, and the y-axis represents the count of customers.
- The chart indicates that most customers have 5 or 6 relationships, with fewer customers having 1-4 relationships.
- The overall distribution suggests that a majority of customers have multiple relationships with the bank.



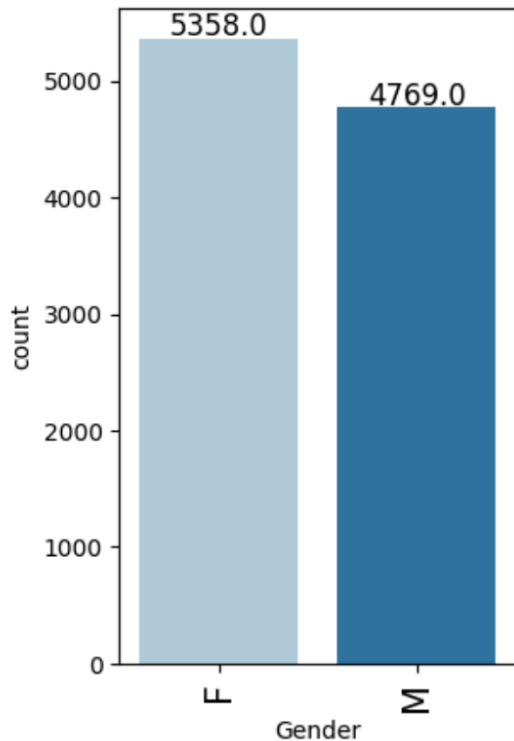
- Marital_Status

- The bar plot shows the distribution of the "Marital_Status" variable.
- The x-axis represents the marital status categories (Divorced, Married, Single), and the y-axis represents the count of customers within each category.
- The chart indicates that most customers are Married, followed by Single and Divorced.
- The distribution suggests a significant majority of customers are married, with a smaller proportion being single or divorced.



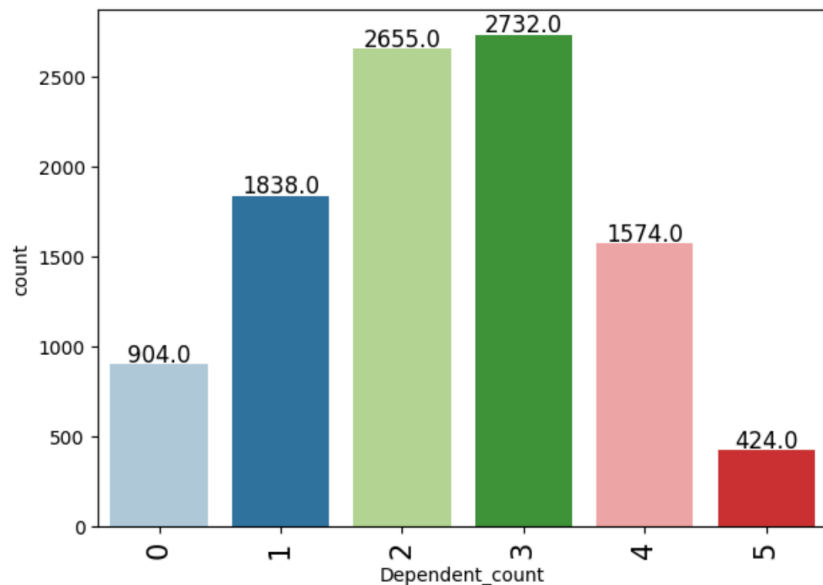
- Income_Category

- The bar plot shows the distribution of the "Income_Category" variable.
- The x-axis represents the income categories, and the y-axis represents the count of customers within each category.
- The chart indicates that most customers have an income between \$40K and \$120K, followed by less than \$40K.
- The distribution suggests a significant portion of customers have higher incomes, while a smaller proportion have lower incomes.



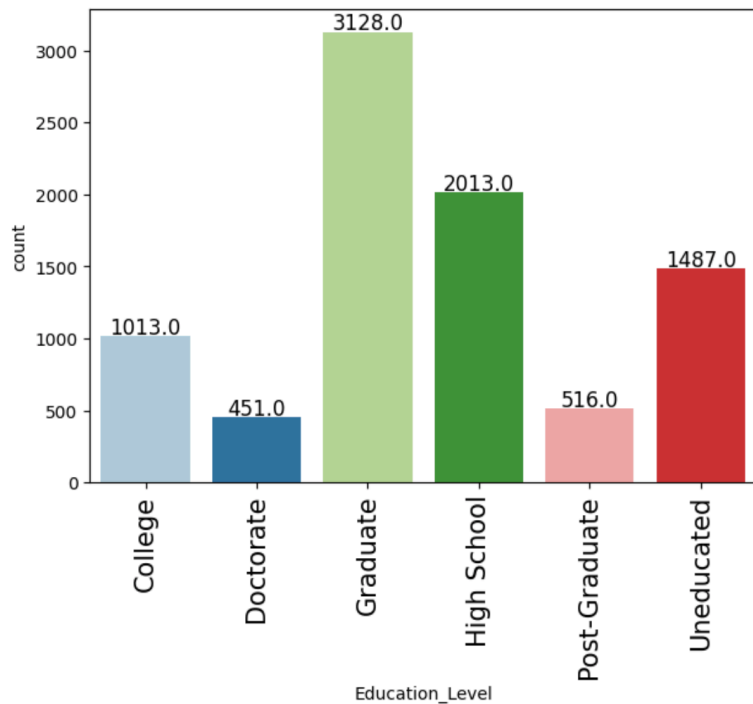
- Gender

- The bar plot shows the distribution of the "Gender" variable.
- The x-axis represents the gender categories (Female, Male), and the y-axis represents the count of customers within each category.
- The chart indicates a slightly higher number of customers are Female compared to Male.
- This suggests a relatively balanced gender distribution in the dataset.



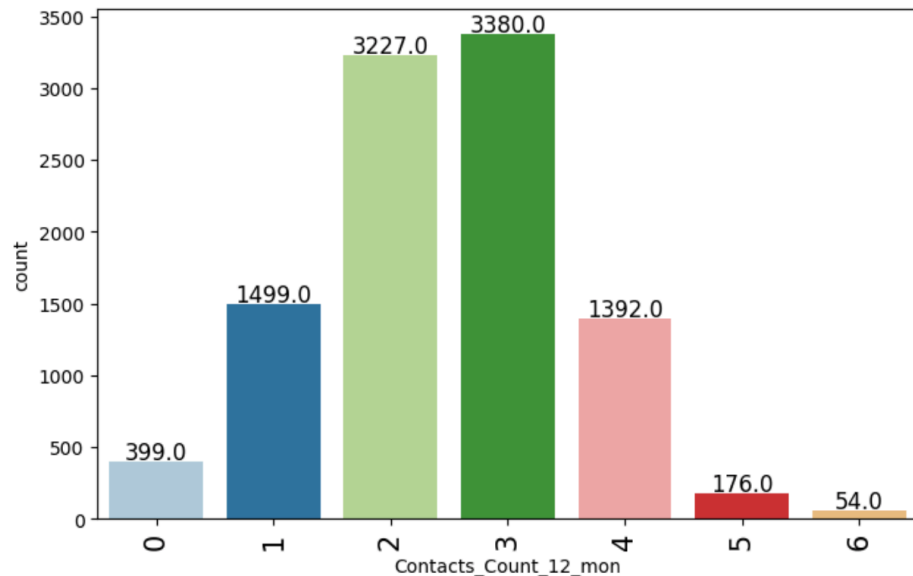
- **Dependent_count**

- The bar plot shows the distribution of the "Dependent_count" variable. The x-axis represents the number of dependents, and the y-axis represents the count of customers within each category.
- The chart indicates that most customers have 3 dependents, followed by 1, 2 and 4 dependents.
- There is a significant drop-off in the number of customers with 5 or more dependents.
- The overall distribution suggests a concentration of customers with 2-3 dependents.



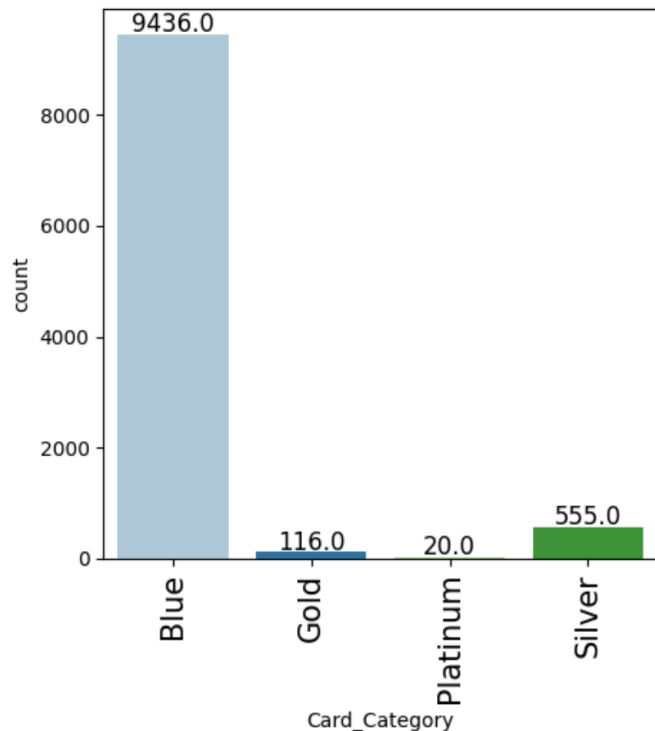
- Education_Level

- The bar plot shows the distribution of the "Education_Level" variable.
- The x-axis represents the education levels, and the y-axis represents the count of students within each level.
- The chart indicates that most students have a Graduate degree, followed by Uneducated and High School.
- The distribution suggests a focus on higher education among the students in the dataset.



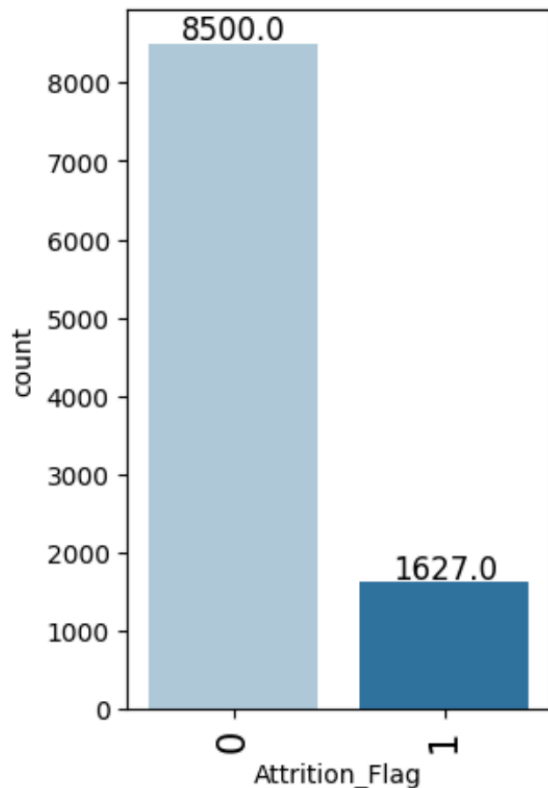
- `Contacts_Count_12_mon`

- The bar plot shows the distribution of the "Contacts_Count_12_mon" variable.
- The x-axis represents the number of contacts made in the past 12 months, and the y-axis represents the count of customers within each category.
- The chart indicates that most customers have had 2-3 contacts in the past 12 months, with a significant drop-off in the number of contacts beyond 4.
- The overall distribution suggests a concentration of customer interactions within the first few months.



- Card_Category

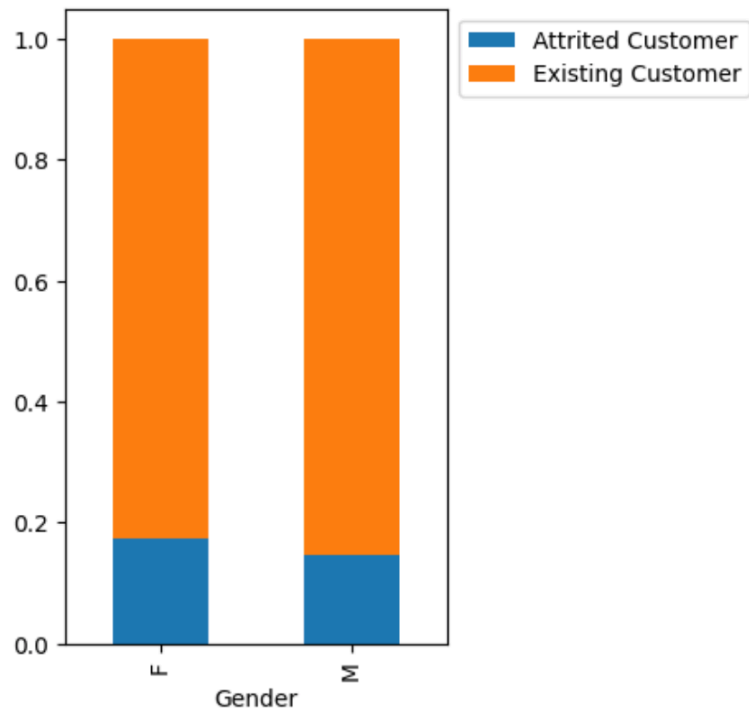
- The bar plot shows the distribution of the "Card_Category" variable.
- The x-axis represents the card categories (Blue, Gold, Platinum, Silver), and the y-axis represents the count of customers within each category.
- The chart indicates that most customers have a Blue card, followed by Silver and Gold.
- The Platinum category has a very low number of customers.
- This suggests that the majority of customers hold basic or mid-tier cards, with a smaller proportion having premium cards.



- Attrition_Flag

- The bar plot shows the distribution of the "Attrition_Flag" variable.
- The x-axis represents the attrition flag (0 for non-attrition, 1 for attrition), and the y-axis represents the count of customers within each category.
- The chart indicates a significant majority of customers have not churned (Attrition_Flag=0), while a smaller proportion have churned (Attrition_Flag=1).
- This suggests that customer retention is relatively high in the dataset.

Attrition_Flag	Attrited Customer	Existing Customer	All
Gender			
All	1627	8500	10127
F	930	4428	5358
M	697	4072	4769



● Attrition_Flag vs Gender

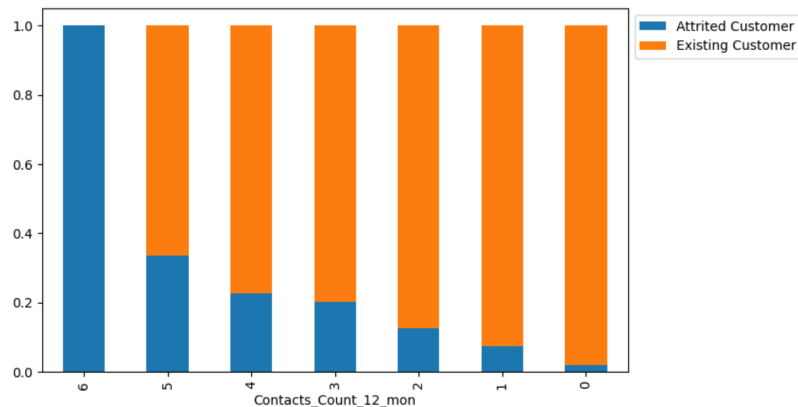
- More female customers have churned compared to male customers.
- The churn rate for both genders is relatively high.

Overall: 1627 customers have Attrition_Flag, while 8500 remain active Existing Customer.

Gender:

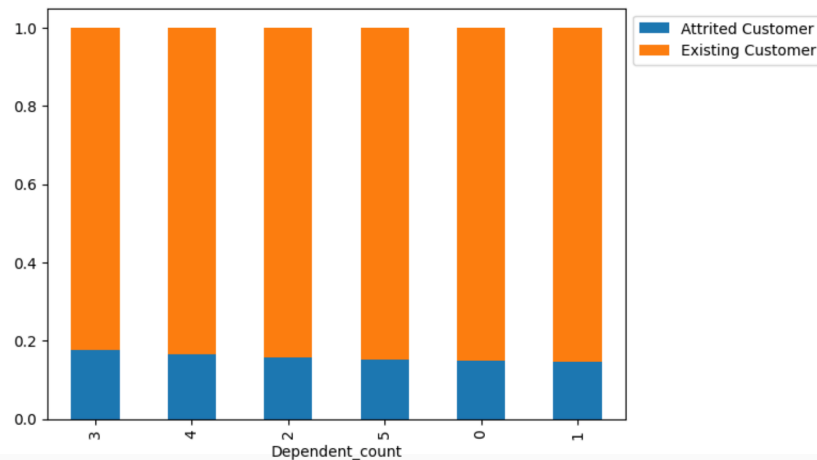
- Female: 930 customers have churned, and 4428 remain active.
- Male: 697 customers have churned, and 4072 remain active.

Attrition_Flag	Attrited Customer	Existing Customer	All
Contacts_Count_12_mon			
All	1627	8500	10127
3	681	2699	3380
2	403	2824	3227
4	315	1077	1392
1	108	1391	1499
5	59	117	176
6	54	0	54
0	7	392	399



- Attrition_Flag vs
Contacts_Count_12_mon
- The chart indicates that most customers have 6 Contacts_Count_12_mon, followed by 5 and 4.
- There is a significant drop-off in the number of customers with 5.
- The overall suggests a concentration of customer interactions within the first few months.

Attrition_Flag	Attrited Customer	Existing Customer	All
Dependent_count			
All	1627	8500	10127
3	482	2250	2732
2	417	2238	2655
1	269	1569	1838
4	260	1314	1574
0	135	769	904
5	64	360	424



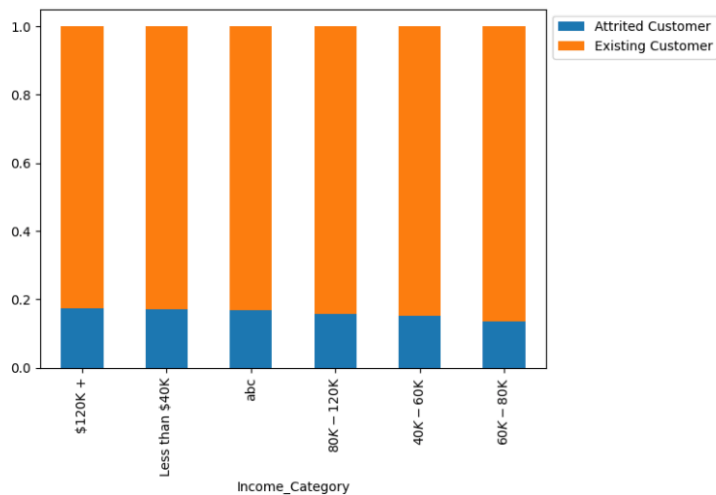
- Attrition_Flag vs Dependent_count

There's not much difference between Dependent_count 0-5.

Attrition_Flag vs Income_Category

```
stacked_barplot(data,"Income_Category", "Attrition_Flag") ## Complete the code to create distributi
```

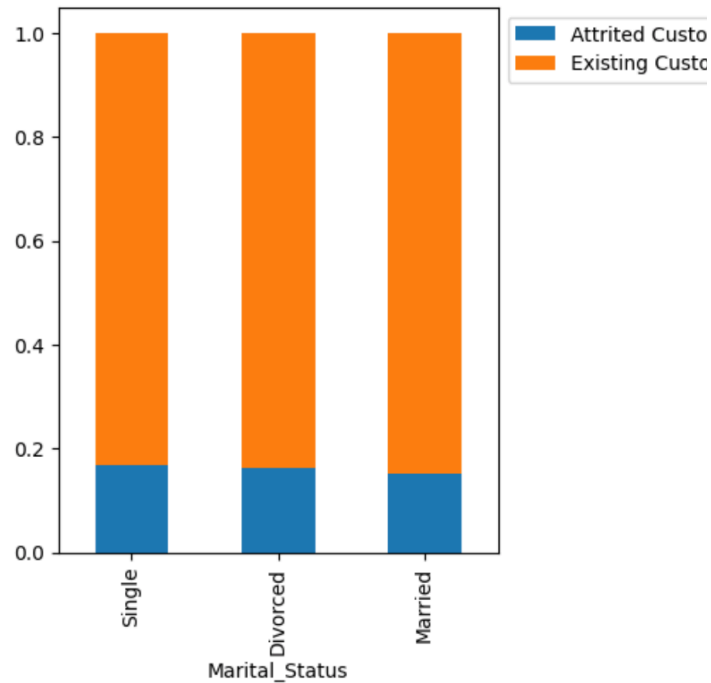
Attrition_Flag	Attrited Customer	Existing Customer	All
Income_Category			
All	1627	8500	10127
Less than \$40K	612	2949	3561
\$40K - \$60K	271	1519	1790
\$80K - \$120K	242	1293	1535
\$60K - \$80K	189	1213	1402
abc	187	925	1112
\$120K +	126	601	727



- Attrition_Flag vs Income_catagory

There's not much difference between Income_catagory.

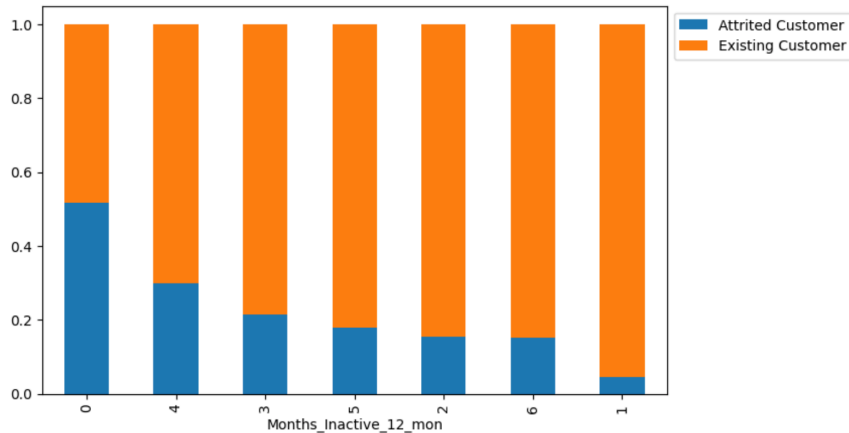
Attrition_Flag	Attrited Customer	Existing Customer	All
Marital_Status			
All	1498	7880	9378
Married	709	3978	4687
Single	668	3275	3943
Divorced	121	627	748



- Attrition_Flag vs Marital_Status

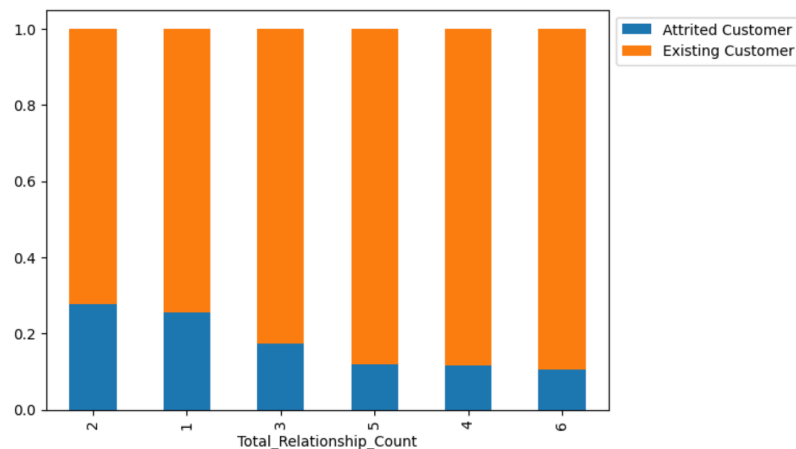
There's not much difference between Marital_Status.

Attrition_Flag	Months_Inactive_12_mon	Attrited Customer	Existing Customer	All
All		1627	8500	10127
3		826	3020	3846
2		505	2777	3282
4		130	305	435
1		100	2133	2233
5		32	146	178
6		19	105	124
0		15	14	29



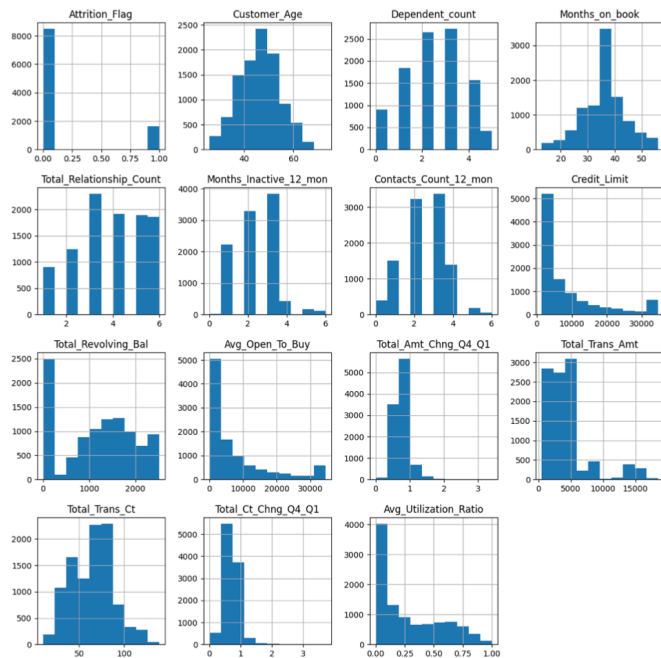
- Attrition_Flag vs Months_Inactive_12_mon
- The chart indicates that most attributed customers have had 0-4 contacts in the past 12 months, with a significant drop-off in the number of contacts beyond 4.

Attrition_Flag	Attrited Customer	Existing Customer	All
Total_Relationship_Count			
All	1627	8500	10127
3	400	1905	2305
2	346	897	1243
1	233	677	910
5	227	1664	1891
4	225	1687	1912
6	196	1670	1866



● Attrition_Flag vs Total_Relationship_Count

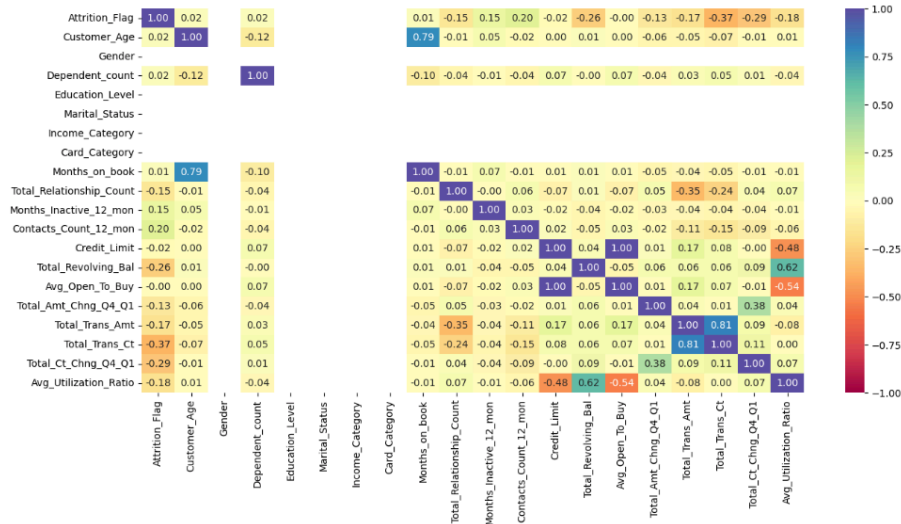
- The number of attrited customers decreases as the number of relationships increases.
- Customers with 1 or 2 relationships have the highest attrited rates.
- Customers with 5 or more relationships have the lowest churn rates.



● Histogram

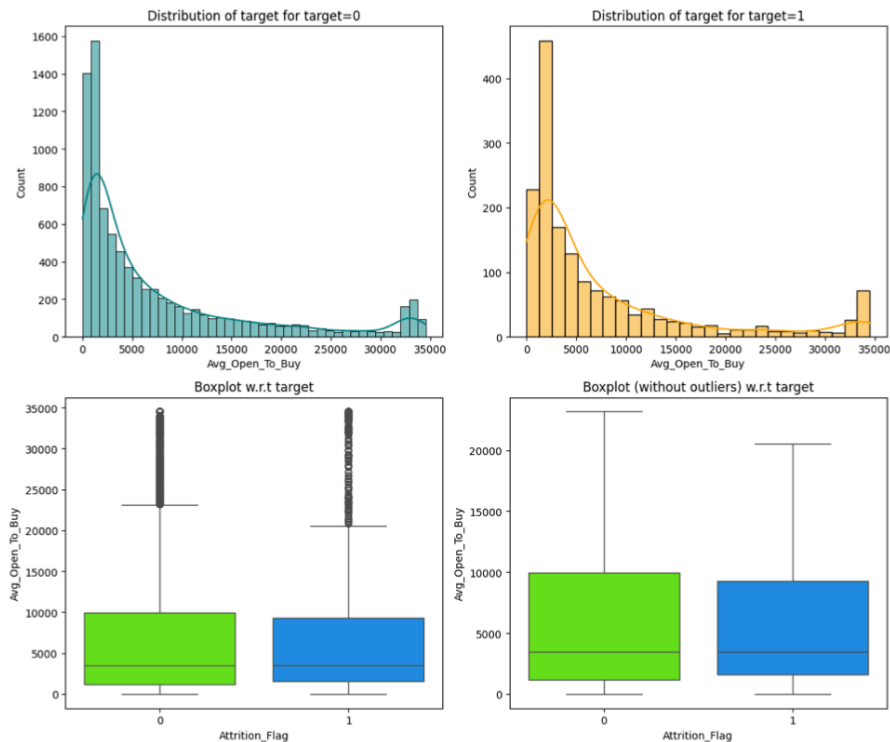
- The image shows histograms for 14 variables.
- The x-axis represents the range of each variable, and the y-axis represents the frequency or count of observations within each bin.
- The histograms provide insights into the distribution of each variable, such as skewness, central tendency, and spread.
- Overall, the visualizations reveal a mix of normally distributed, skewed, and categorical variables, providing valuable information about the characteristics of the dataset.

14



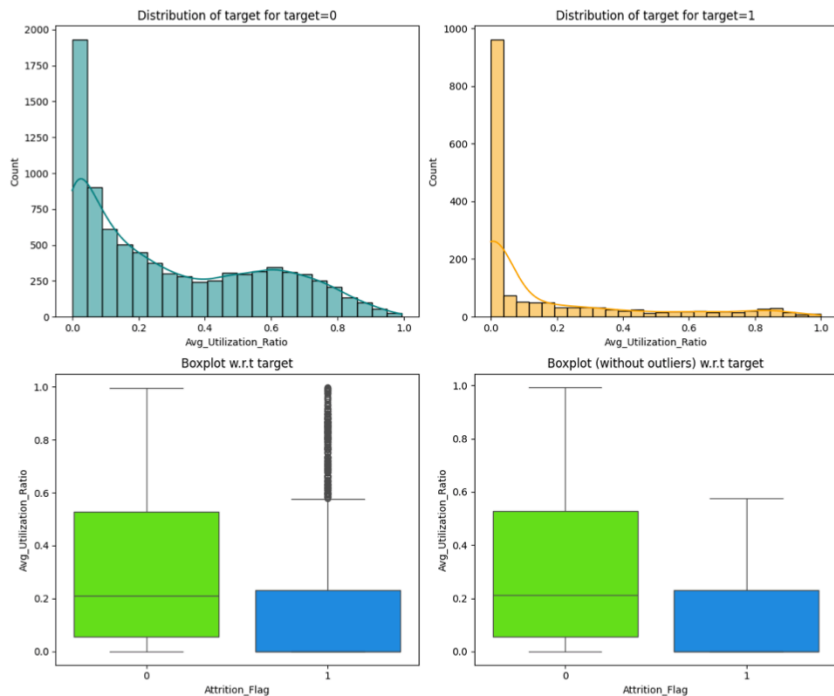
Correlation Check

- The diagonal line shows the perfect correlation of each variable with itself (1.00).
- By examining the correlation matrix, you can identify variables that are highly correlated or have little or no correlation with each other.



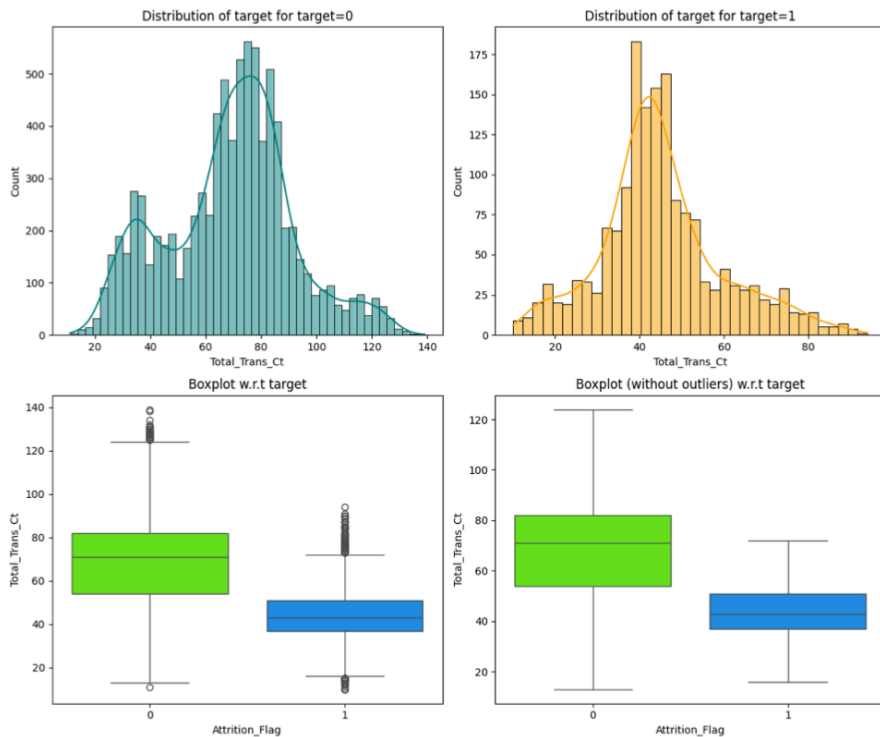
● Attrition_Flag vs Avg_Open_To_Buy

- The image shows box plots of "Avg_Utilization_Ratio" for churned (target=1) and non-churned (target=0) customers.
- The distributions reveal that customers who churned tend to have lower average open to buy compared to those who did not churn.
- The box plots further illustrate this difference, with the median and quartiles of the "Attrition_Flag" variable being lower for the churned group.



● Avg_Utilization_Ratio vs Attrition_Flag

- The image shows the distribution of the "Avg_Utilization_Ratio" variable for churned (target=1) and non-churned (target=0) customers.
- The distribution for churned customers is skewed to the right, indicating higher average utilization ratios, while the distribution for non-churned customers is more concentrated on lower values.
- The box plots further illustrate this difference, with the median and quartiles of the "Attrition_Flag" variable being lower for non-churned (target=0) group.

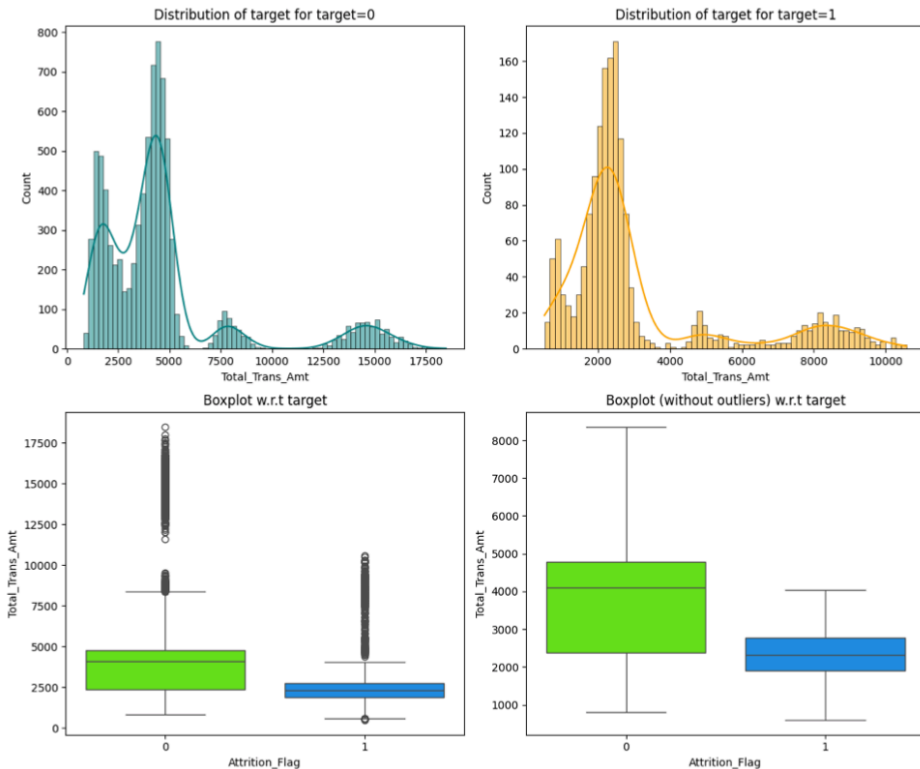


● Total_Trans_Ct vs Attrition_Flag

The image shows the distribution of "Total_Trans_Ct" for churned (target=1) and non-churned (target=0) customers.

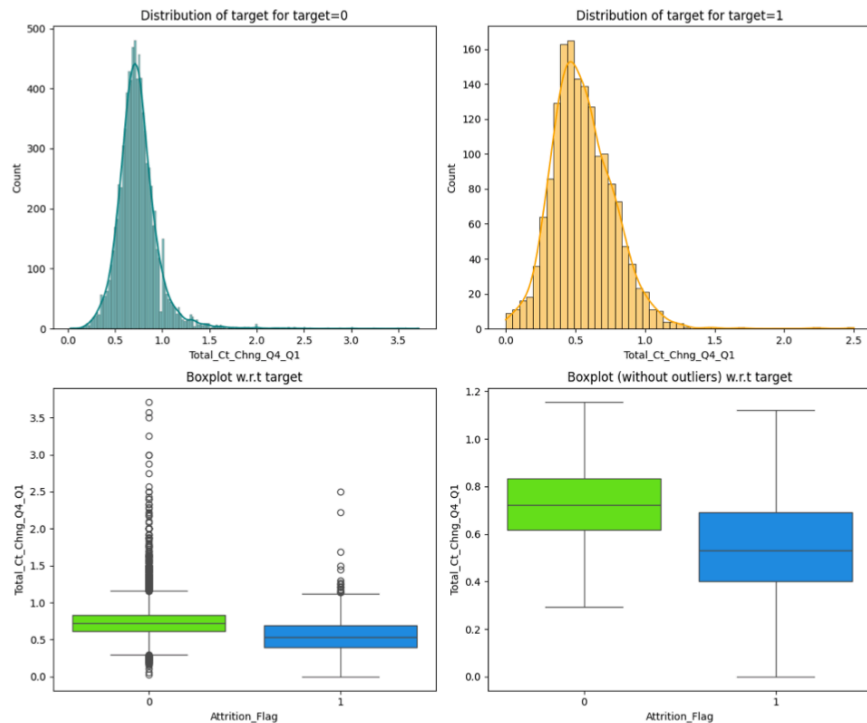
The distributions reveal that churned customers tend to have lower total transaction counts compared to non-churned customers.

The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the churned (target=1) group.



• Total_Trans_Amt vs Attrition_Flag

- The image shows the distribution of "Total_Trans_Amt" for churned (target=1) and non-churned (target=0) customers.
- The distributions reveal that churned customers tend to have lower total transaction amounts compared to non-churned customers.
- The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the churned (target=1) group.

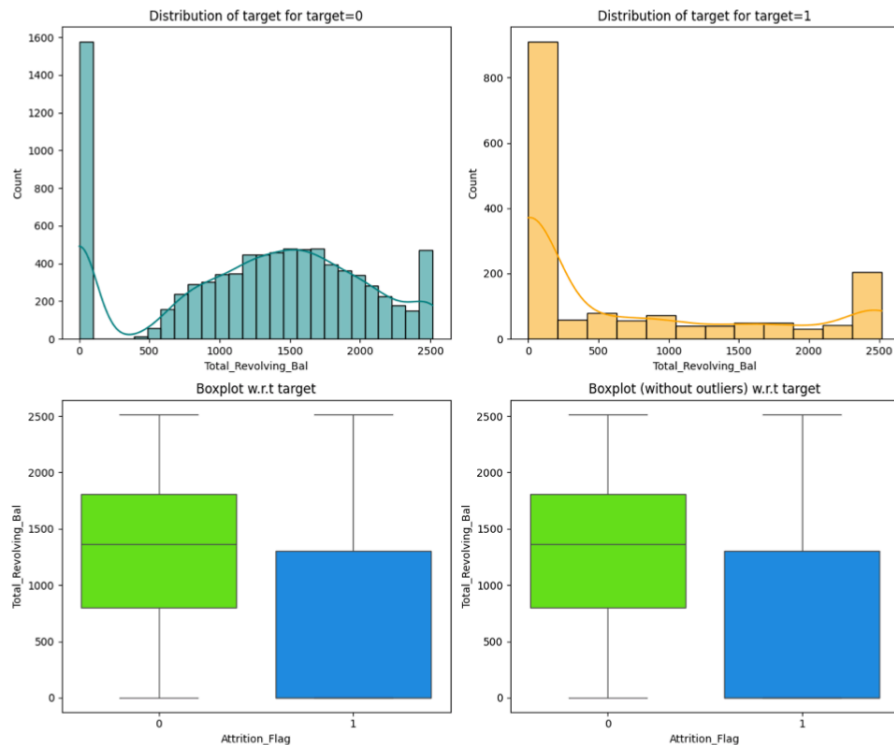


● Total_Ct_Chng_Q4_Q1 vs Attrition_Flag

The image shows the distribution of "Total_Ct_Chng_Q4_Q1" for churned (target=1) and non-churned (target=0) customers.

The distributions reveal that churned customers tend to have lower total count changes compared to non-churned customers.

The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the non-churned (target=0) group.

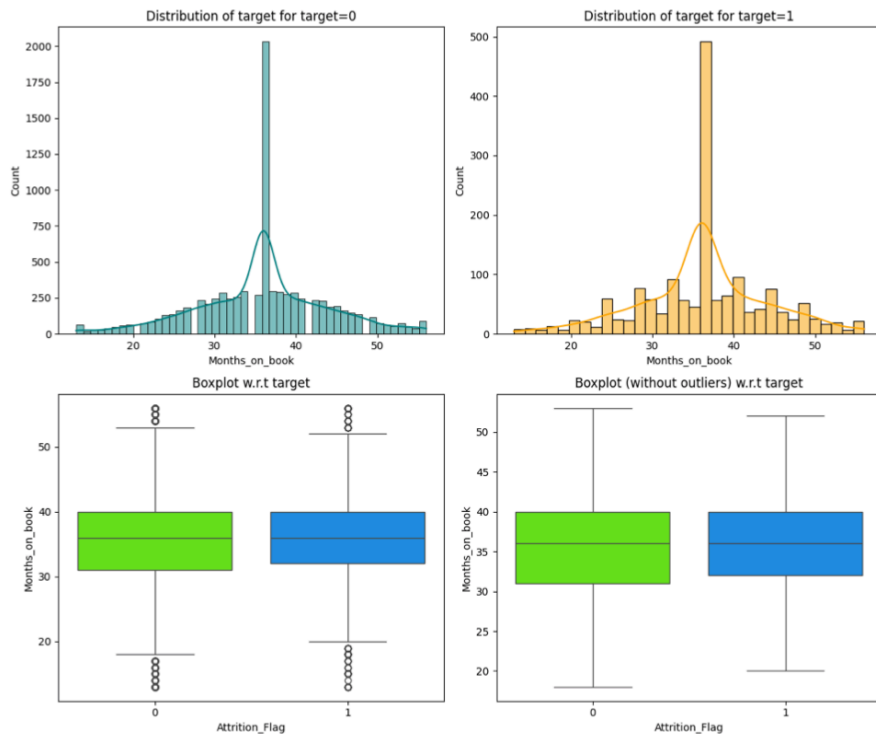


● Attrition_Flag vs Total_Revolving_Bal

The image shows the distribution of "Total_Revolving_Bal" for churned (target=1) and non-churned (target=0) customers.

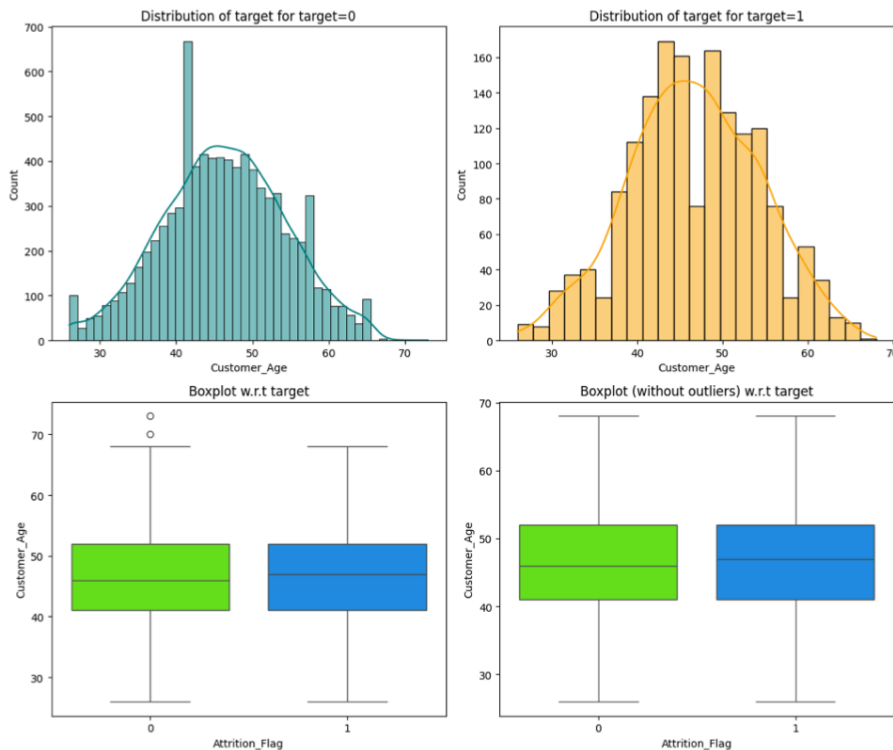
The distributions reveal that churned customers tend to have lower total revolving balances compared to non-churned customers.

The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the churned (target=1) group..



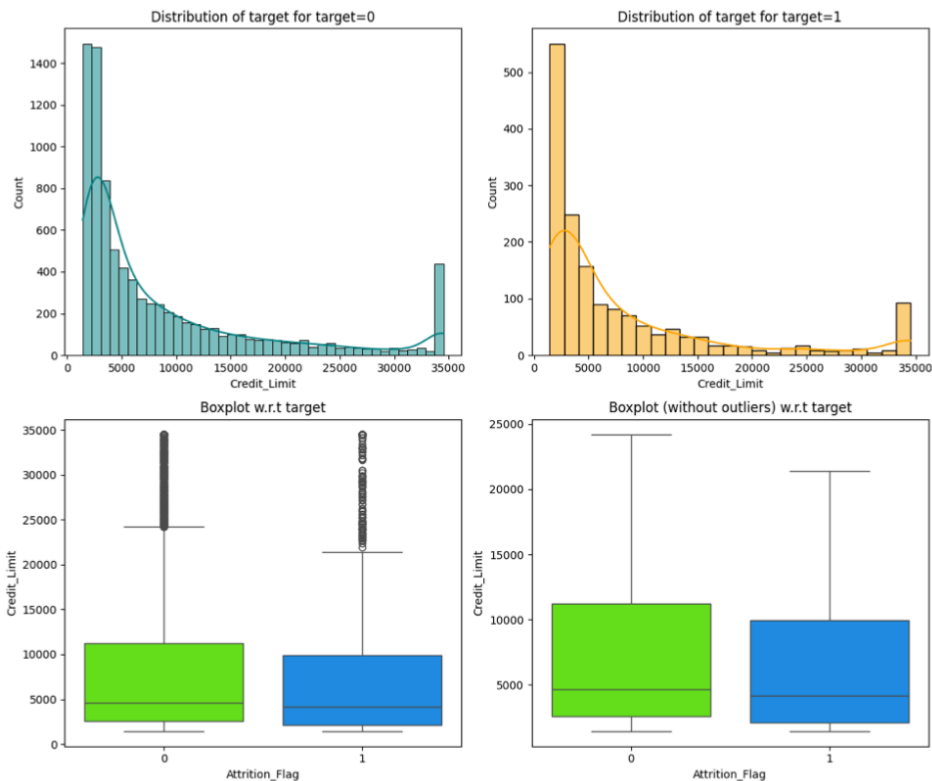
● Attrition_Flag vs Months_on_book

- The image shows the distribution of "Months_on_book" for churned (target=1) and non-churned (target=0) customers.
- The distributions reveal that churned customers tend to have lower tenure with the bank compared to non-churned customers.
- The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the churned group.



● Attrition_Flag vs Customer_Age

- The image shows the distribution of "Customer_Age" for churned (target=1) and non-churned (target=0) customers.
- The distributions reveal that churned customers tend to be slightly younger than non-churned customers.
- The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" are not much different.



● Attrition_Flag vs Credit_Limit

- The image shows the distribution of "Credit_Limit" for churned (target=1) and non-churned (target=0) customers.
- The distributions reveal that churned customers tend to have lower credit limits compared to non-churned customers.
- The box plots further illustrate this difference, with the median and quartiles of "Attrition_Flag" being lower for the churned group.

Data Preprocessing

- Duplicate value check

There are no duplicate values in the data.

- Missing value treatment

There are 1,519 missing values in Education_Level and 749 in Marital_Status.

	0
Attrition_Flag	16.066
Customer_Age	0.020
Gender	0.000
Dependent_count	0.000
Education_Level	0.000
Marital_Status	0.000
Income_Category	0.000
Card_Category	0.000
Months_on_book	3.812
Total_Relationship_Count	0.000
Months_Inactive_12_mon	3.268
Contacts_Count_12_mon	6.211
Credit_Limit	9.717
Total_Revolving_Bal	0.000
Avg_Open_To_Buy	9.509
Total_Amt_Chng_Q4_Q1	3.910
Total_Trans_Amt	8.848
Total_Trans_Ct	0.020
Total_Ct_Chng_Q4_Q1	3.891
Avg_Utilization_Ratio	0.000

- Outlier check

Most important features: The top 5 features with the highest importance scores are:

Attrition_Fla Months_on_book (16.066)

Total_Trans_Ct (0.020)

Total_Amt_Chng_Q4_Q1 (3.891)

Avg_Open_To_Buy (9.509)

Credit_Limit (9.717)

Least important features: The remaining features have an importance score of 0.000, indicating that they have little or no impact on the model's predictions. These results suggest that the length of time customers have been with the bank (Months_on_book), the number of transactions (Total_Trans_Ct and Total_Trans_Amt), the credit limit and available credit (Credit_Limit and Avg_Open_To_Buy), and changes in spending patterns (Total_Amt_Chng_Q4_Q1) are the most important factors influencing customer churn.

- **Feature engineering**

- Unique values in Gender are 5,358 female customers and 4,769 male customers in the dataset.
- Unique values in Education_Level are 3,128 customers with a Graduate degree, 2,013 with a High School diploma, 1,487 with no education, 1,013 with a College degree, 516 with a Post-Graduate degree, and 451 with a Doctorate.
- Unique values in Marital_Status are 4,687 married customers, 3,943 single customers, and 748 divorced customers.
- Unique values in Income_Category are 3,561 customers with an income less than \$40K, 1,790 customers with an income between \$40K and \$60K, 1,535 customers with an income between \$80K and \$120K, 1,402 customers with an income between \$60K and \$80K, 1,112 customers with an unknown income category (possibly due to missing or incorrect data), and 727 customers with an income greater than \$120K.
- Unique values in Card_Category are 9,436 Blue cardholders, 555 Silver cardholders, 116 Gold cardholders, and 20 Platinum cardholders.

- **Data preparation for modeling**

- The shape of the training set (X_train) has 8101 samples and 19 features
- The shape of the validation set (X_val) has 507 samples and 19 features
- The shape of the test set (X_test) has 1519 samples and 19 features.
- There is no missing value training and validation sets.
- There are 232 missing values in the Education_Level column and 127 missing values in the Marital_Status column of the test set.

- Checking inverse mapped values/categories.
- **Training Data**
 - There are 4,279 female customers and 3,822 male customers in Gender.
 - There are 3,733 customers with a Graduate degree, 1,619 with a High School diploma, 1,171 with Uneducated, 816 with a College degree, 407 with a Post-Graduate degree, and 355 with a Doctorate degree in Education_Level.
 - There are 4,346 married customers, 3,144 single customers, and 611 divorced customers Marital_Status.
 - There are 2,812 customers with an income less than \$40K, 1,453 customers with an income between \$40K and \$60K, 1,237 customers with an income between \$80K and \$120K, 1,122 customers with an income between \$60K and \$80K, 889 customers with an unknown income category, and 588 customers with an income greater than \$120K in Income_Category.
 - There are 7,557 Blue cardholders, 436 Silver cardholders, 93 Gold cardholders, and 15 Platinum cardholders in Card_Category.

- **Validation Data**

- There are 266 female customers and 241 male customers in Gender.
- There are 237 customers with a Graduate degree, 94 with a High School, 84 with Uneducated, 49 with a College, 19 with a Post-Graduate degree, and 24 with a Doctorate degree in Education_Level.
- There are 272 married customers, 193 single customers, and 42 divorced customers Marital_Status.
- There are 174 customers with an income less than \$40K, 88 customers with an income between \$40K and \$60K, 71 customers with an income between \$80K and \$120K, 74 customers with an income between \$60K and \$80K, 62 customers with an unknown income category, and 38 customers with an income greater than \$120K in Income_Category.
- There are 465 Blue cardholders, 37 Silver cardholders, 3 Gold cardholders, and 2 Platinum cardholders in Card_Category.

- **Testing Data**

- There are 4271 female customers and 3822 male customers in Gender.
- There are 3733 customers with a Graduate degree, 1619 with a High School, 1171 with Uneducated, 816 with a College, 407 with a Post-Graduate degree, and 355 with a Doctorate degree in Education_Level.
- There are 4346 married customers, 3144 single customers, and 611 divorced customers Marital_Status.
- There are 2812 customers with an income less than \$40K, 1453 customers with an income between \$40K and \$60K, 1237 customers with an income between \$80K and \$120K, 1122 customers with an income between \$60K and \$80K, 889 customers with an unknown income category, and 588 customers with an income greater than \$120K in Income_Category.
- There are 7557 Blue cardholders, 436 Silver cardholders, 93 Gold cardholders, and 15 Platinum cardholders in Card_Category.

- Encoding categorical variables

After encoding the categorical variables, the shape of the training set (X_{train}) is (8101, 30), the shape of the validation set (X_{val}) is (507, 30), and the shape of the test set (X_{test}) is (1519, 30).

Model Performance Summary (original data)

- Summary of performance metrics for training and validation data in tabular format for comparison for original data.

The training model performances:

- **Random Forest and dtree:** These models achieved 1.0 on the training set.
- **Bagging:** Achieved 0.978.
- **Adaboost:** Achieved 0.871.
- **GBM:** Achieved 0.894.

The validation model performances:

- **Bagging and Adaboost:** These models achieved the highest performance on the validation set, with 0.851.
- **Random Forest:** Achieved 0.743.
- **dtree:** Achieved 0.824.
- **GBM:** Achieved 0.851.

- Comments on the model performance (original data)

Training model performances

Random Forest and dtree appear to be the top-performing models on the training set. However, it's important to evaluate the models on a separate test set to assess their generalization ability and avoid overfitting.

The validation model performances:

Bagging and Adaboost appear to be the top-performing models on the validation set. However, it's important to consider the following:

Model Performance Summary (oversampled data)

- The oversampling method

Before Oversampling, counts of label 'Yes': 1300

Before Oversampling, counts of label 'No': 6801

After Oversampling, counts of label 'Yes': 6801

After Oversampling, counts of label 'No': 6801

After Oversampling, the shape of train_X: (13602, 30)

After Oversampling, the shape of train_y: (13602,)

- **Summary of performance metrics for training and validation data in tabular format for comparison for oversampled data.**
- Training Performance:
 - **GBM, and dtree:** These models achieved the highest performance with approximately 0.905.
 - **Bagging:** Achieved of 0.892.
 - **Random Forest:** Achieved of 0.838.
 - **Adaboost:** Achieved of 0.811.
- Validation Performance:
 - **GBM, and dtree:** These models achieved the highest performance on the validation set, with approximately 0.905.
 - **Bagging:** Achieved of 0.892.
 - **Random Forest:** Achieved of 0.838.
 - **Adaboost:** Achieved of 0.811.

- **Comments on the model performance (oversampled data)**

- Training Performance:

GBM, and dtree appear to be the top-performing models on the training data. However, it's important to evaluate the models on a separate test set to assess their generalization ability and avoid overfitting.

- Validation Performance:

GBM, and dtree appear to be the top-performing models on the validation set as well. This suggests that they are able to generalize well to unseen data and might be suitable for making predictions on new customers.

Model Performance Summary (undersampled data)

- The undersampling method

Before Under Sampling, counts of label 'Yes': 1300

Before Under Sampling, counts of label 'No': 6801

After Under Sampling, counts of label 'Yes': 1300

After Under Sampling, counts of label 'No': 1300

After Under Sampling, the shape of train_X: (2600, 30)

After Under Sampling, the shape of train_y: (2600,)

- **Summary of performance metrics for training and validation data in tabular format for comparison for undersampled data.**

- Training Performance:

GBM: Achieved approximately 0.959.

Bagging and dtree: Achieved 0.905 and 0.919, respectively.

Random Forest: Achieved 0.919.

Adaboost: Achieved 0.932.

- Validation Performance:

GBM: Achieved approximately 0.959.

Bagging and dtree: Achieved 0.905 and 0.919, respectively.

Random Forest: Achieved 0.919.

Adaboost: Achieved 0.932.

- Comments on the model performance (undersampled data)

- Training Performance:

Overall, GBM appear to be the top-performing models on the training data. However, it's important to evaluate the models on a separate test set to assess their generalization ability and avoid overfitting.

- Validation Performance:

Overall, GBM appear to be the top-performing models on the validation set as well. This suggests that they are able to generalize well to unseen data and might be suitable for making predictions on new customers.

Model performance summary for hyperparameter tuning

- **Tuning AdaBoost using original data**

After careful hyperparameter tuning, we determined the optimal model configuration with the following parameters: `n_estimators=100`, `learning_rate=0.1`, and `estimator=DecisionTreeClassifier(max_depth=3, random_state=1)`. This model achieved a remarkable cross-validation score of 0.87, indicating exceptional predictive performance.

The performance on the training set is Accuracy 0.985, Recall 0.934, Precision 0.969, F1 0.951.

The performance on the validation set is Accuracy 0.966, Recall 0.892, Precision 0.88, F1 0.886.

- **Tuning Ada Boost using undersampled data**

The performance on the training set is Accuracy 0.992, Recall 0.993, Precision 0.991, F1 0.992.

The performance on the validation set is Accuracy 0.937, Recall 0.946, Precision 0.714, F1 0.814.

- **Tuning Gradient Boosting using undersampled data**

Through extensive hyperparameter tuning, we identified the optimal model configuration with the following parameters: subsample=0.9, n_estimators=100, max_features=0.5, learning_rate=0.1, and init=AdaBoostClassifier(random_state=1). This model achieved a cross-validation score of 0.8392, demonstrating strong predictive performance.

The performance on the training set is Accuracy 0.968, Recall 0.971, Precision 0.966, F1 0.968.

The performance on the validation set is Accuracy 0.929, Recall 0.932, Precision 0.690, F1 0.793.

- **Tuning Gradient Boosting using original data**

Through meticulous hyperparameter tuning, we identified the optimal model configuration with the following parameters: subsample=0.9, n_estimators=100, max_features=0.5, learning_rate=0.1, and init=AdaBoostClassifier(random_state=1). This model achieved a cross-validation score of 0.8392, demonstrating robust predictive performance.

The performance on the training set is Accuracy 0.975, Recall 0.881, Precision 0.959, F1 0.918.

The performance on the validation set is Accuracy 0.975, Recall 0.881, Precision 0.959, F1 0.918.

Tuning Gradient Boosting using over sampled data

The performance on the training set is Accuracy 0.954, Recall 0.962, Precision 0.946, F1 0.954.

The performance on the validation set is Accuracy 0.929, Recall 0.932, Precision 0.690, F1 0.793.

Tuning XGBoost Model with Original data

Through rigorous hyperparameter tuning, we identified the optimal model configuration with the following parameters: subsample=0.7, scale_pos_weight=5, n_estimators=75, learning_rate=0.05, and gamma=3. This model achieved a remarkable cross-validation score of 0.9346, indicating exceptional predictive performance.

The performance on the training set is Accuracy 0.949, Recall 0.971, Precision 0.772, F1 0.860.

The performance on the validation set is Accuracy 0.929, Recall 0.932, Precision 0.690, F1 0.793.

Model Comparison and Final Model Selection

Training performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.968	0.954	0.992
Recall	0.971	0.962	0.993
Precision	0.966	0.946	0.991
F1	0.968	0.954	0.992

validation performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.929	0.929	0.937
Recall	0.932	0.932	0.946
Precision	0.690	0.690	0.714
F1	0.793	0.793	0.814

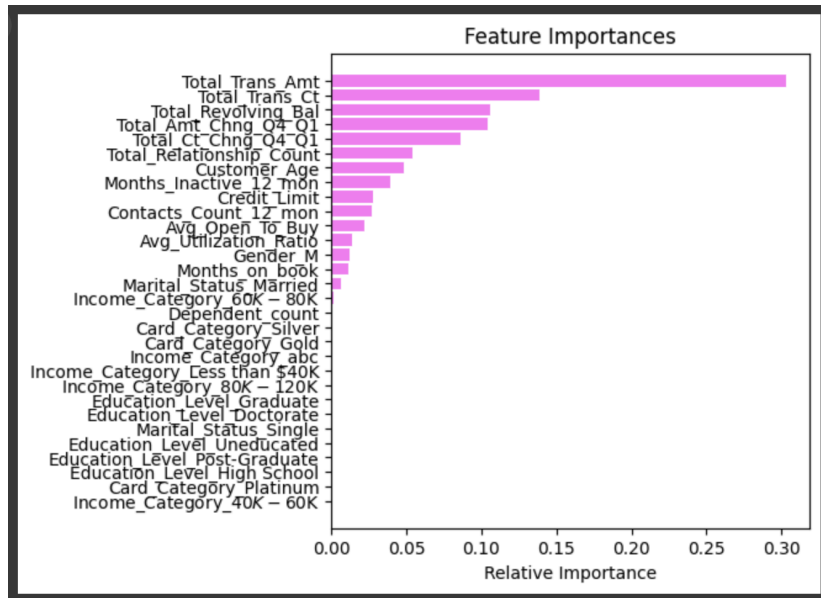
Overall, Adaboost trained with Undersampled data achieved the highest performance across all metrics, indicating its effectiveness in handling the imbalanced dataset. Gradient Boosting models also performed well, with similar results regardless of whether the data was undersampled or not. This suggests that both Gradient Boosting and Adaboost are suitable for this classification task, and the choice of undersampling might not have a significant impact on the final performance.

Final model is performing on unseen test data

XGBoost Model with Test data

The performance on the validation set is Accuracy 0.967, Recall 0.846, Precision 0.951, F1 0.895.

Feature Importances



Most important features: The top 5 features with the highest importance scores are:

- Total_Trans_Amt
- Total_Trans_Ct
- Total_Revolving_Bal
- Total_Amt_Chng_Q4_Q1
- Total_Ct_Chng_Q4_Q1

Least important features: The remaining features have an importance score of 0.00, indicating that they have little or no impact on the model's predictions.

Business Insights and Conclusions

- The analysis reveals that several factors are associated with customer churn, including credit limit, available credit, spending patterns, and tenure with the bank. By addressing these factors, such as offering higher credit limits, improving customer engagement, and providing incentives for long-term customers, the bank can effectively reduce churn and improve customer retention.

APPENDIX



Happy Learning !

